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Interpretable prediction of short-term load curve for extreme scenarios based on mixture of experts model

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ABSTRACT

The current short-term load forecasting models face challenges such as low accuracy under extreme weather conditions and holidays, and a lack of transparency in the forecasting process, which leads to poor reliability and interpretability. Therefore, combined with explainable artificial intelligence, an extreme scenario explainable load forecasting method based on a mixture of experts model is proposed. The total load is deconstructed into multiple explainable components and predicted by multiple expert models. Firstly, the conventional daily load components were extracted by combining meteorology with the clustering method. Then, the mapping relationship between influencing factors and different load components is established using prior knowledge, and the mixture of experts (MoE) is improved. According to the coupling relationship of different load components, the explainable model structure and learning objectives are constructed by prior enablement. Finally, according to the load characteristics of different components, eXtreme Gradient Boosting (XGBoost) and Informer were used to establish corresponding expert models, respectively, and the total load prediction results were obtained by adding and reconstructing the load prediction results of different components. Simulation results show that the proposed method can effectively improve prediction accuracy under extreme weather, holidays, and other scenarios, and the proposed method has good interpretability.

1. Introduction

Load forecasting is critically important in the power industry as it directly impacts system operation, dispatch, planning, and market efficiency. Accurate forecasting not only enhances grid performance and optimizes resource allocation but also provides key information for the market, helping to establish reasonable pricing mechanisms, ensure reliable power delivery, improve system operation levels, and guide long-term infrastructure planning [1]. At present, load forecasting can be divided into long-term, medium-term, short-term, and ultra-short-term forecasting, according to different forecasting cycles [2]. Among these, short-term forecasting primarily focuses on load variations from one day to one week ahead [3]. The results are often directly utilized for real-time dispatch decisions, such as power generation planning and unit commitment optimization [4]. The accuracy of these forecasts is crucial for the safe, stable, and economical operation of the power system. However, due to the inherent characteristics of short-term power load and external factors such as weather, holidays, and user behavior, it exhibits strong randomness and complexity. Moreover, with the ongoing changes in the global climate system, the frequency and intensity of extreme weather events have been on the rise. During such extreme events, including holidays, heat waves, or severe cold spells, power loads can undergo sudden shifts, disrupting the power system's generation schedule and leading to imbalances between supply and demand [5]. For example, in 2020, regions like Hunan and Jiangxi in China experienced power shortages due to an insufficient supply during a cold wave. Similarly, in 2022, Sichuan Province faced both power and energy deficits caused by extreme weather conditions such as unusually high temperatures and drought.

Traditional short-term load forecasting methods include the Autoregressive Model [6] (AR), Auto Regressive Moving Average Model (ARMA), and Auto-Regressive Integrated Moving Average Model [7] (ARIMA), etc. Any of these methods are based on linear assumptions, which assume that the relationship between the load and time or other factors is linear, whereas, in practice, the load usually exhibits non-linear characteristics. Machine learning methods mainly include

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0378-7796/© 2025 Elsevier B.V. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

Decision Trees, Random Forest [8], and Gradient Boosting Trees. However, machine learning models can easily overfit training data, especially when there is insufficient data or the model complexity is too high. With the continuous development of artificial intelligence, some deep learning methods are widely used, such as Recurrent Neural Network [9](RNN), Long Short-Term Memory network [10](LSTM), Gated Recurrent Unit (GRU), etc.

Due to the complexity of load sequences, load forecasting remains challenging when relying on a single algorithm. Recently, signal decomposition techniques have gained popularity in load forecasting. Literature [11] proposed a hybrid model combining variable mode decomposition (VMD) and LSTM, which removes seasonal factors and corrects errors. Literature [12] introduced a method using empirical mode decomposition (EMD) and bi-directional LSTM, which decomposes time series into components with varying resolutions, improving forecasting accuracy. These methods primarily focus on decomposing high- and low-frequency components of the load sequence to enhance accuracy during stable periods. However, they overlook the physical meaning of electric power load and are purely data-driven, lacking correlations with influencing factors and interpretability. Consequently, the prediction models established under these conditions may exhibit certain drawbacks. When meteorological factors undergo unexpected changes or holidays approach, the accuracy of load forecasting can significantly decline. This is because, under extreme scenarios, the power load demonstrates stronger volatility and abruptness, particularly during sudden temperature drops or heat waves. In addition, most of the current literature considers load forecasting under general weather conditions [13,14], and few scholars take the impact of extreme weather into account when making short-term load forecasting, and still face significant difficulties in special scenarios such as holidays and extreme weather.

In the traditional short-term load forecasting modeling process, although load analysis is the predecessor task of forecasting, the results of the load characteristics obtained from the analysis are often not effectively inputted into the forecasting model, and there is a lack of effective methods to inject expert knowledge into the model [15]. Therefore, there is an urgent need to develop a load forecasting model that can incorporate the knowledge of experts and is interpretable. However, the existing research on the interpretability of load forecasting models is mainly through correlation analysis, time series decomposition, and attribution interpretation [16]. The correlation analysis mainly calculates the correlation coefficient between the power load and the influencing factors and uses the thermal map to visualize the relationship between the variables and the load. Most of the current time series decomposition methods directly apply signal processing algorithms to decompose load or trend and seasonal components, etc. Such methods lack expert knowledge injection, and their interpretability is difficult to convince. For example, literature [17] proposes improved Fourier transform and seasonal decomposition methods to forecast data after trend decomposition, which improves accuracy and interpretability to a certain extent. Attributional interpretation interprets the results of the model after training, such as SHAP and LIME algorithms. For example, in literature [18], the SHAP method was used to explain the feature importance of load forecasting models after model training. Literature [19] proposes an approach based on the local interpretable model-agnostic explanations (LIME) method, which computes the local feature contribution after the predictive model is trained so as to better understand the obtained results. The above algorithm is characterized by no effect on model performance and is usually a model-independent method [20], which is usually untargeted. Explainable artificial intelligence (XAI) provides an important idea for solving the above problems. Literature [21] gives the basic concept of explainable AI, i.e., explainability is the degree of human understanding of the reasons for a decision, which is intended to enhance the user's understanding and trust in the algorithm. Explanations can be full or partial. In recent years, the mixture of experts [22] (MoE) has been widely used in the field of artificial intelligence. By effectively using the advantages of multiple expert models, MoE not only improves model performance and computational efficiency but also enhances model flexibility and interpretability. Due to the relative independence of the expert models within the mixture of experts, its decision-making process is more transparent and easier to understand compared to traditional models. This high interpretability is of significant importance for application fields that require clear explanations of the model's reasoning logic.

This paper applies the improved MoE to power load forecasting in this context. It proposes an interpretable method for short-term load forecasting in extreme scenarios based on a mixture of experts. The total load can be decomposed into multiple interpretable components, which multiple expert models can predict. Major contributions are as follows:

(1) A novel classification framework for regional loads is developed. By combining the pentad mean temperature method with K-Medoids clustering, this approach accurately categorizes regional loads and extracts conventional daily load patterns. The resulting classification provides robust data support for interpretable load decomposition, particularly in distinguishing temperature-sensitive components from baseline demand.

(2) A novel load decomposition method is proposed, which differs from traditional signal processing approaches by integrating prior knowledge into the model architecture. This method establishes causal mappings between influencing factors (e.g., temperature, holidays) and specific load components. Training prediction models based on the coupling relationships among these components effectively decompose complex nonlinear relationships into multiple interpretable load components, including conventional load, temperature-regulating load, and holiday load.

(3) A new load forecasting framework is proposed, combining expert models with prior knowledge integration. It constructs interpretable model structures and learning objectives before training, decomposing the original load into distinct components. Establishing expert prediction models for each component significantly improves forecasting performance under extreme weather, holidays, and other scenarios while enhancing the interpretability of both the model and its predictions.

The remainder of this paper is organized as follows: Section 2 describes the adaptive division method for routine daily load. Section 3 details the proposed framework, including model improvements, feature selection, expert model configuration, and evaluation metrics. Section 4 introduces the experimental setup, dataset, and validation results. Finally, Section 5 presents the conclusion and discusses directions for future work.

2. Adaptive daily load extraction method based on meteorology and curve clustering

In load forecasting under extreme scenarios, it is crucial to accurately separate the regular load component from load fluctuations caused by special events such as extreme weather or holidays. By extracting typical regular-day load patterns, temperature-sensitive load can then be effectively distinguished from them, providing a solid foundation for the subsequent application of specialized forecasting models.

Firstly, the spring and fall time periods containing regular days are first divided based on the meteorological method of seasonal dynamics, which typically relies on calendar months or ambient temperature criteria. In this paper, we adopt an absolute temperature threshold-based method for seasonal classification: when the temperature at the ground station exceeds (or falls below) a certain threshold, it is defined as the beginning (or end) of a particular season [23]. This approach is used in the operational definition of the four seasons by the China Meteorological Administration, according to the national standard "Climate Seasonal Classification" implemented in 2023. If the

average temperature at a ground station remains above 22 °C for five consecutive days, it is defined as summer; if it remains below 10 °C, it is defined as winter; and if the average temperature is between 10 and 22 °C, it is defined as spring or fall.

$$P_i = \frac{T_{i-4} + T_{i-3} + T_{i-2} + T_{i-1} + T_i}{5} \tag{1}$$

where: P_i is the migratory temperature on day i; T_i is the average air temperature on day i.

Determining regular days based solely on seasonal division is imprecise; further screening using curve shape is necessary. This study proposes a method combining cosine similarity and the K-Medoids clustering algorithm [24] to analyze spring daily loads. Based on the clustering results, time periods with temperature regulation loads in spring and fall are excluded, and seasonal division nodes are adjusted for the adaptive extraction of conventional daily load periods.

Let $H_i = [h_{i1}, h_{i2}, \ldots, h_{iM}]$ be the load data series of day i, $H_j = [h_{j1}, h_{j2}, \ldots, h_{jM}]$ is the load data sequence on day j, and M is the load sequence number of daily sampling points, then the cosine similarity between H_i and H_j is calculated as:

$$S_{\cos}(H_i, H_j) = \frac{H_i H_j}{|H_i||H_j|} = \frac{\sum_{n=1}^{M} h_{in} h_{jn}}{\sqrt{\sum_{n=1}^{M} h_{in}^2}\sqrt{\sum_{n=1}^{M} h_{jn}^2}} \tag{2}$$

In the above equation, the cosine similarity S_{cos} takes the value interval of [0,1]. When the value of S_{cos} tends to be closer to 1, it indicates that the similarity of the load curve shapes represented by the load data series H_i and H_j on the day i and day j are higher.

To find the optimal number of clusters, it is necessary to evaluate the validity of the clustering results. In this paper, the contour coefficient $s(i)$ and the sum of squares of errors are used as the assessment indices. The calculation process of contour coefficient $s(i)$ and error squared is as follows:

The contour coefficients of each point within the sample are first calculated and then averaged to obtain the contour coefficients of the whole sample. For the i-th point in the sample, the formula for calculating the contour coefficient is as follows:

$$s(i) = \frac{b(i) - a(i)}{\max\{a(i), b(i)\}} \tag{3}$$

where $a(i)$ denotes the average cosine similarity of sample H_i with other samples H_j within the same cluster, while $b(i)$ denotes the average cosine similarity of sample H_i with all other samples H_j in other clusters. By calculating the average of the profile coefficients of all samples, the overall profile coefficient S of the clustering result can be obtained.

$$S = \frac{1}{N}\sum_{i=1}^{N} s(i) \tag{4}$$

where the overall profile coefficient S takes values in the range of $[-1,1]$, the closer the value of S is to 1 the better the clustering effect.

The sum of squared errors of the samples S_{SSE} is calculated as:

$$S_{\mathrm{SSE}} = \sum_{i=1}^{n} \sum_{p \in C_i} |p - m_i|^2 \tag{5}$$

where n is the total number of sample points and C_i is the training dataset; p is a sample point in C_i and m_i is the center of mass of C_i.

3. Short-term load curves for extreme scenarios based on the mixture of experts can explain the forecasting method

3.1. Improved mixture of experts

This paper enhances load forecasting accuracy and robustness with an improved mixture of experts model [22]. As shown in Fig. 1, residual linkage structures are added to better separate and model load components, improving the network's predictive ability by decoupling and independently processing these components.

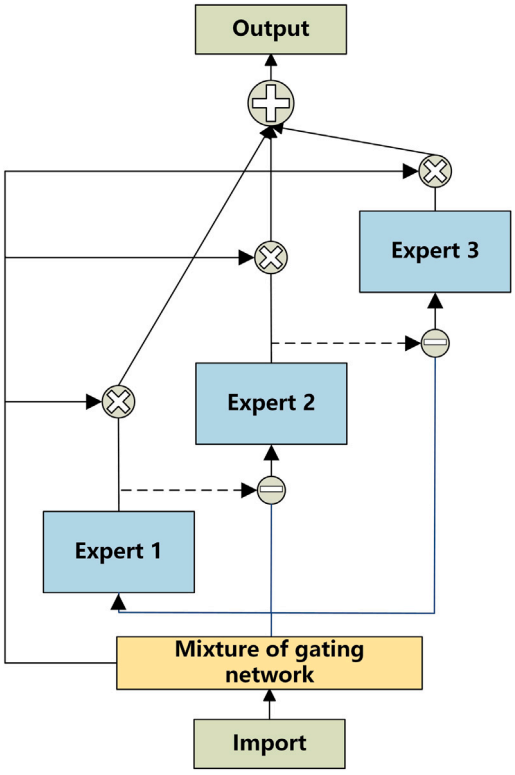


Fig. 1. Improved mixture of experts' network model structure diagram.

The expression of the mixture of experts model is shown as follows.

$$y = E(x) \cdot G(x) = \sum_{i}^{n} e_i(x)\, g_i(x) \tag{6}$$

where $E(x)$ is the output of the expert network, $G(x)$ is the output of the gated network(GateNet), $e_i(x)$ and $g_i(x)$ then correspond to the outputs of the$i$th position of the expert and the gated network, respectively, and the gated network and the expert receive the same inputs. The gated network divides the single-task space into multiple sub-tasks. Then, multiple expert networks (sub-models) process the specific sub-tasks separately to obtain the overall prediction result. In this paper, we use a priori empowerment to construct an interpretable model structure and learning objectives and fix the allocation rules of the mixture of experts' gated networks, which uses deterministic decision-making to assign specific tasks to selected experts explicitly. Hence, the gated network described in this paper is a pre-defined allocation rule.

This paper decomposes the original electric load curve into regular, temperature-regulating, and holiday load components using a mixture of experts model. This approach leverages known components to predict unknown ones through multi-component deconstruction. The load exhibits stable patterns in normal conditions without heating, cooling, or holiday effects. High or low temperatures add temperature-regulating loads, while holidays introduce holiday load components. In some cases, all three components may overlap, requiring further analysis based on load characteristics. Based on the above analysis, $G(T)$, $H(T)$, $J(T)$, and $K(T)$ are set to represent the total load, regular load, thermoregulation load, and holiday load components, respectively, for the period T and holiday load components. T_1, T_2, T_3, and T_4 are coupled time periods for different components, then there are:

$$\begin{aligned} G(T_1) &= H(T_1) \\ G(T_2) &= H(T_2) + J(T_2) \\ G(T_3) &= H(T_3) + J(T_3) + K(T_3) \\ G(T_4) &= H(T_4) + K(T_4) \end{aligned} \tag{7}$$

Can be solved for each load component:

$$\begin{aligned} H(T_1) &= G(T_1) \\ J(T_2) &= G(T_2) - H(T_2) \\ K(T_3) &= G(T_3) - H(T_3) - J(T_3) \\ K(T_4) &= G(T_4) - H(T_4) \end{aligned} \tag{8}$$

Therefore, the expert prediction models can be trained for $H(T)$, $J(T)$, and $K(T)$ respectively based on the coupling relationship of Eq. (8). The prediction results of different expert models can be summed up to obtain$x$moment of the total load prediction results, as shown in Eq. (9).

$$G(x) = H(x) + J(x) + K(x) \tag{9}$$

This paper assumes that the total load consists of a linear superposition of multiple independent components, thus decomposing the original data into different data subspaces, so the weight of each expert model of the gated network is always 1.

3.2. Classification of influencing factors based on causation

This paper constructs a causal map based on the causal relationship between events and combines a priori knowledge and expert experience to classify the influencing factors. Causality defines the objective link between "cause" and "effect" events, where the former triggers the latter [25]. To explore causality from the data perspective, Pearl proposed a structural causal model (SCM) [26], which describes the causal mechanism through a set of equations, and an SCM generally consists of the following equations:

$$x_i = f_i(\mathrm{pa}_i, u_i), i = 1, 2, \cdots, n \tag{10}$$

where: pa_i denotes the variable, x_i is the set of parent variables, and u_i denotes the error caused by the omitted factor.

Thus, a complete causal relationship usually includes the following three elements: the cause variable pa_i, the effect variable x_i, and the causal mechanism f_i. To facilitate reasoning over the graph, the causal graph is formally represented as $D=(V,C)$, where $V=\{v_1, v_2,\ldots, v_n\}$ is the set of nodes in the graph, v_i represents events, and $C=\{c_1, c_2, \ldots, c_m\}$ represents events and the set of relationships between events, c_i represents the causal relationship between events and events. The specific process is as follows: first, based on a priori knowledge and expert experience, to obtain the cause of changes in different load components of the event. For example, meteorological changes (causal event) → changes in users' demand for temperature comfort (causal event) → changes in the frequency of use of air conditioners, water heaters, and other thermostatic devices (causal event) → changes in the thermostatic load. Next, the mapping relationship between the cause event and the variables is constructed to establish a complete causal map. For instance, "meteorological changes" correspond to variables like humidity, temperature, and pressure; "holiday effects" correspond to variables like holiday type and day; and "changes in equipment use" map to intraday variations. Finally, causal nodes triggering load component changes are traversed, and relevant variables are aggregated to form the complete set of influencing factors. The inference algorithm is detailed below:

(1) Initialize the set of influencing factors, initialize the node access flags, input the graph structure corresponding to the graph $D=(V,C)$, and take the event node describing a change in the load component as the start node;

(2) The access to the set of nodes $V=\{v_1, v_2,\ldots, v_n\}$starting from the node where the load component has changed, the nodes are visited sequentially in the opposite direction of the influence factor conduction;

(3) If the currently visited node v_i exists a neighboring node that has not yet been visited, select one at random to visit, and if the node does not have any nodes accessible from it, add it to the set of influencing factors and return the previously visited node that still has unvisited neighbors;

(4) Repeat until all nodes connected to the start node have been visited, returning the set of influencing factors.

3.3. Selection of expert models of different load components

To predict the regular and holiday load components, which mainly vary with time, XGBoost [27] is used as the regression model. The cumulative effect of temperature over time must be considered for the temperature regulation load, which is influenced by meteorological factors and exhibits a lag correlation with temperature. Thus, Informer [28], a time series model based on the Transformer architecture, is selected to capture these long-term dependencies through its Self-attention mechanism.

3.4. Overall process of extreme scenario explainable load forecasting based on the mixture of experts model

The coupling of power consumption habits, climate, holidays, and other factors affects the actual power consumption characteristics, resulting in prominent non-stationary characteristics of the data. Therefore, this paper divides the electric load data into three components: regular load, temperature-regulated load, and holiday load, and the overall framework of the extreme scenario explainable load prediction model based on the mixture of experts model is shown in Fig. 2.

(1) Data Preprocessing and Feature Engineering — Includes feature construction, missing value filling, and outlier repair.

(2) Regular Day Time Extraction — This method uses meteorological methods to analyze temperature dynamics, clusters spring and fall load curves via K-Medoids, excludes periods with temperature-regulated loads based on evaluation indices, and extracts regular daily load components.

(3) Causal Analysis and Feature Mapping — Combines prior knowledge and expert insights to identify factors affecting load variations, establishing mappings between influencing factors and load components.

(4) Model Construction and Prediction — Introduces residual connections into a mixture-of-experts framework for modeling coupled load components. XGBoost predicts regular and holiday loads, while Informer handles temperature-regulated loads. Steps include:

First, the holidays are stripped, and a regular load component prediction model is built based on the accessible regular daily load to predict the regular load component of the complete training set.

Second, the regular load component is obtained by subtracting the regular load component from the temperature-regulating daily load and building a prediction model, and then the temperature-regulating load component is predicted for the holiday period.

Finally, the holiday load component is obtained by subtracting the conventional load component and the temperature-regulating load component from the holiday load and establishing a prediction model. Finally, the prediction model of different load components is obtained.

(5) Final prediction and interpretation — Based on the mapping between the load components and the factors that influence the process, the relevant features are input into separate expert models. The predictions for each load component are combined to generate the final total load prediction, which can be further interpreted through interpretable components.

3.5. Model evaluation index

The root mean square error (RMSE), the mean absolute error (MAE), and the mean absolute percentage error (MAPE) were chosen as the model evaluation indices.

$$RMSE = \sqrt{\frac{1}{n}\sum_{i=1}^{n}\left(y_i - \hat{y}_i\right)^2} \tag{11}$$

$$MAE = \frac{1}{n}\sum_{i=1}^{n} \mid \hat{y}_i - y_i \mid \tag{12}$$

$$MAPE = \frac{100\%}{n}\sum_{i=1}^{n} \mid \frac{\hat{y}_i - y_i}{y_i} \mid \tag{13}$$

In the above evaluation index calculation formula, y_i is the actual observed value, $\hat{y}_i$ is the model predicted value, and n is the number of samples.

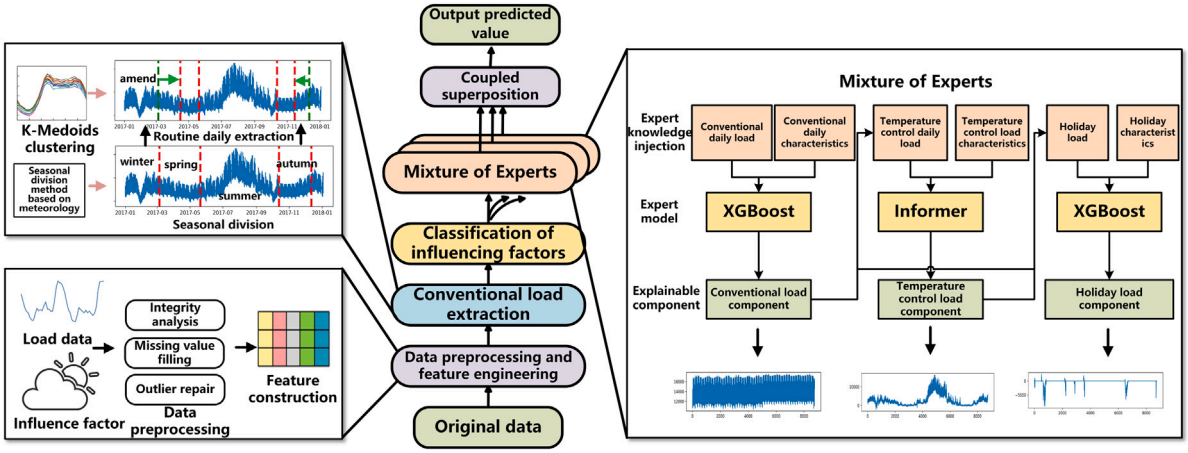


Fig. 2. Prediction flow chart based on MoE.

Table 1
Results of seasonal division based on meteorology.

Season	Spring	Summer	Autumn	Winter
Division Results	2017/3/5	2017/5/20	2017/10/13	2017/12/12

Table 2
Results of routine day time division.

Season	Routine spring days	Routine autumn days
Routine Period	2017/4/15–2017/5/20	2017/10/13–2017/11/15

4. Experiment and analysis

4.1. Data description and data preprocessing

The experiment uses hourly load and meteorological data for a region in eastern China from January 1, 2017, to December 31, 2019. Meteorological data includes temperature, humidity, body temperature, wind speed, wind level, and wind direction. The data is split into a training set (2017 to June 30, 2018), a validation set (July 1 to December 31, 2018), and a test set (2019).

Experiments were conducted on a computer with an AMD Ryzen 5 5600 6-Core processor, NVIDIA GeForce RTX 4070 graphics card, and 16 GB RAM. The software was implemented in Python 3.11.4 using PyTorch and Scikit-learn.

In data preprocessing, anomalies were treated as missing values. Small gaps were filled via linear interpolation, while more significant gaps were filled with data from the same time in the previous year. The initial missing/anomaly rate was 0.06%, with no missing values or outliers remaining. All input features were normalized to the range [0, 1] via linear transformation. Meteorological data were sourced from authoritative forecasts, focusing on utilizing these forecasts rather than predicting the weather.

4.2. Analysis of regular load adaptive extraction results

The seasonal division for the 2017 test set was calculated using the meteorological five-day smoothed series and the marquee mean temperature method. The division nodes for the four seasons are shown in Table 1.

For the analysis of different clustered curves, as shown in Figs. 3 and 4 for the clustering results of the spring and fall curves, in which the first category of curves is much lower than the two or three categories of curves and the date of the curves is weekends, the third category of curves is close to the node of seasonal division, and the overall higher than the second category of curves, which belongs to curves that contain thermoregulation loads.

Therefore, the first category is defined as the non-working day load curve, the second category is the regular working day load curve, and the third category is the existence of the thermoregulation load curve. According to the thermoregulation load curve results, the seasonal division node is corrected to obtain the regular non-cooling load period.

According to the clustering results, the spring and fall seasons are excluded to include the temperature-regulating load periods, and the regular daily load periods are obtained as shown in Table 2.

4.3. Comparative analysis of prediction accuracy of different models

To evaluate the practical effect of the proposed model in the short-term load forecasting task, the proposed forecasting model is compared to the current mainstream forecasting models XGBoost, LSTM [10], Informer and TCN [29] and the VMD-TCN model plus decomposition forecast proposed in the literature [30,31]. The forecasting task is to forecast the load profile 24 h before the forecasting day, with a total of 24-point forecasting values.

To ensure consistency, all models undergo identical preprocessing and feature engineering (It does not include the load decomposition part that divides the power load data into the conventional load, temperature control load, and holiday load). Optimal parameters are determined through a grid search by defining search ranges and step sizes and conducting exhaustive experiments to minimize errors. Input features and parameter settings for the proposed model are detailed in Table 3, with unspecified parameters set to default values. Date features are directly constructed, while meteorological features are derived from processed weather forecasts.

For deep learning prediction models, MSE is used as the loss function. To prevent overfitting, early stopping, and Dropout are applied. Early stopping monitors validation loss at each training cycle and halts training if the loss fails to decrease for three consecutive cycles, preserving the best model. Dropout, set at 0.05 based on experiments, randomly drops neurons during training to enhance regularization while maintaining learning ability. The Adam optimizer is employed to prevent gradient explosion. Prediction results across scenarios are summarized in Tables 4–6 and Fig. 5.

From Table 4, deep learning algorithms demonstrate higher prediction accuracy than traditional machine learning methods on normal days(No holidays, no extreme weather). Informer achieves the lowest weekday MAPE (4.50%), and TCN performs best on weekends (2.51%). The MAPE of this paper's algorithm is 2.91% and 2.73% on weekdays and weekends, respectively, which is 51.42%, 61.66%, 35.33%,

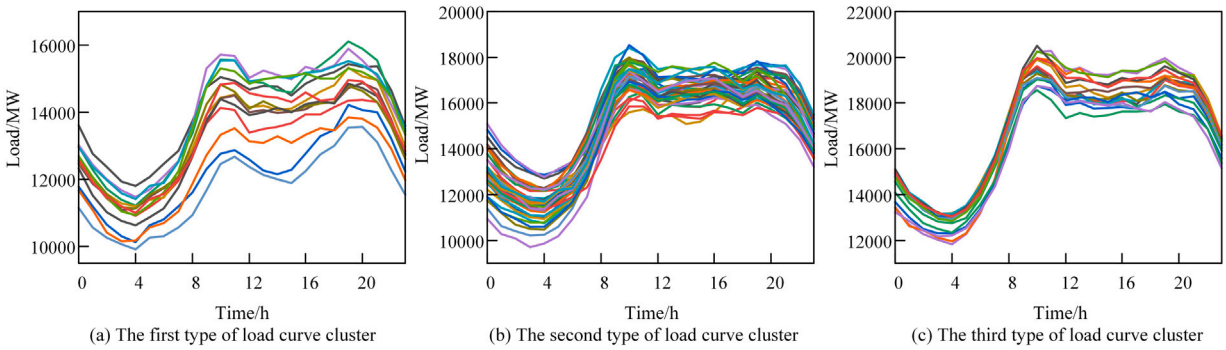


Fig. 3. Spring curve clustering results.

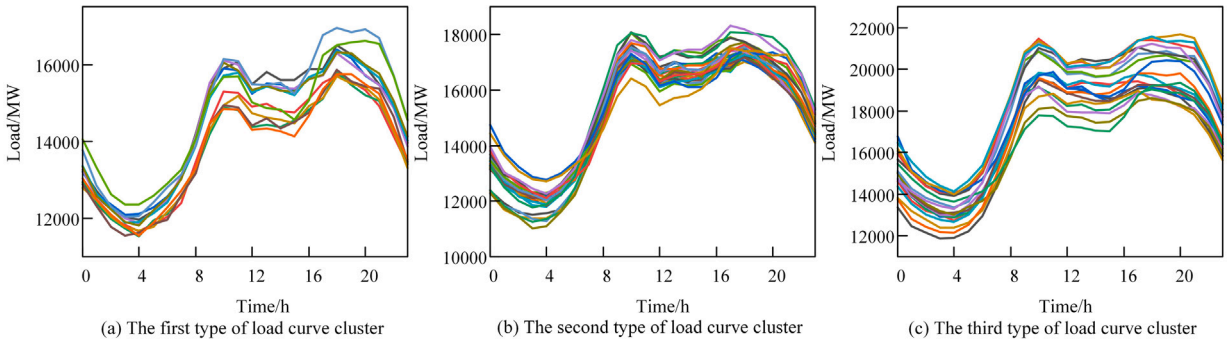


Fig. 4. Autumn curve clustering results.

Table 3
Model Parameters and Input Features.

Model	Parameter settings	Input features
Temperature-Adjusted Load Component Prediction Model (Informer)	seq_len: 24 × 4 label_len: 24 pred_len: 24 encoder input length: 8 decoder input length: 8 predicted output length: 1 learning rate: 0.0001 batch size: 32 dropout rate: 0.05 training epoch: 20 patience: 3 optimizer: Adam	temperature (raw data) humidity (raw data) apparent temperature (raw data) wind speed (raw data) wind level (label encoding) wind direction (label encoding) workday indicator (0–1) hour (1–24)
Regular Load Component Prediction Model (XGBoost)	max_depth: 6 learning rate: 0.1 n_estimators: 300 min_child_weight: 2	year (raw data) month (1–12) workday indicator (0–1) hour (1–24) weekend indicator (0–1)
Holiday Load Component Prediction Model (XGBoost)	max_depth: 5 learning rate: 0.1 n_estimators: 300 min_child_weight: 2	holiday indicator (0–1) holiday name (label encoding) shift adjustment (0–1) hour (1–24) holiday day (1–7)

Table 4
Prediction accuracy comparison for normal days.

Model	Workday (2019-3-11–2019-3-15)			Weekend (2019-3-9–2019-3-10)		
	RMSE/MW	MAE/MW	MAPE/%	RMSE/MW	MAE/MW	MAPE/%
XGBoost	1,107.09	1,025.36	5.99	1,472.91	1,405.20	8.08
LSTM	1,754.30	1,351.90	7.59	1,680.56	1,490.78	8.50
Informer	1,076.48	801.08	4.50	726.02	630.05	3.62
TCN	1,120.71	889.08	4.92	667.52	445.89	2.51
VMD-TCN	1,378.21	1,151.65	4.59	853.88	750.76	4.60
Proposed	**644.93**	**510.25**	**2.91**	**698.93**	**489.61**	**2.73**

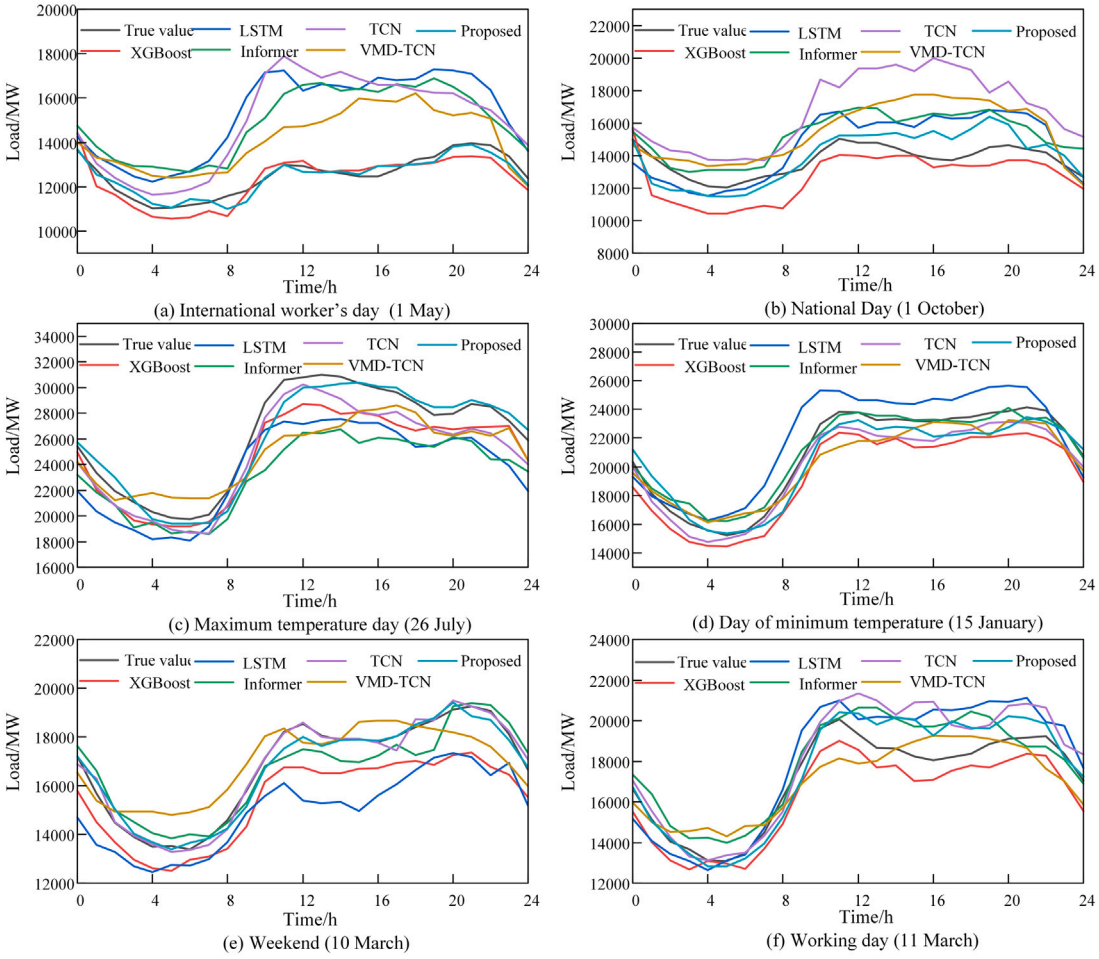


Fig. 5. Prediction curves under different scenarios by different methods.

Table 5
Prediction accuracy comparison for high and low temperature periods.

Model	High temperature (2019-07-22—2019-08-12)			Low temperature (2019-01-13—2019-01-27)		
	RMSE/MW	MAE/MW	MAPE/%	RMSE/MW	MAE/MW	MAPE/%
XGBoost	1,858.51	1,616.56	6.73	1,294.01	1,135.09	5.72
LSTM	1,964.03	1,667.14	6.88	1,833.16	1,494.79	7.51
Informer	2,013.03	1,639.97	6.63	843.61	641.12	3.22
TCN	1,459.77	1,187.93	5.10	902.47	689.50	3.41
VMD-TCN	1,700.31	1,370.02	5.86	1,018.10	840.65	4.26
Proposed	**1,190.00**	**919.55**	**3.88**	**754.54**	**616.45**	**3.12**

Table 6
Prediction accuracy comparison for holidays.

Model	National Day (2019-10-1)			International Worker's Day (2019-5-1)		
	RMSE/MW	MAE/MW	MAPE/%	RMSE/MW	MAE/MW	MAPE/%
XGBoost	1,268.90	1,105.18	8.29	460.03	400.52	3.23
LSTM	1,588.91	1,359.33	9.73	3,108.01	2,818.12	22.38
Informer	1,709.80	1,475.28	10.60	2,639.14	2,374.14	18.92
TCN	3,542.60	3,149.60	22.66	2,927.53	2,460.73	19.45
VMD-TCN	2,165.60	1,849.51	13.34	1,924.48	1,726.89	13.85
Proposed	**960.08**	**797.47**	**5.81**	**271.93**	**222.21**	**1.79**

40.85%, and 36.60% lower in weekday scenarios compared to XGBoost, LSTM, Informer, TCN, and VMD-TCN in weekend scenarios, LSTM, Informer, and VMD-TCN in weekend scenarios by 66.21%, 67.88%, 24.59%, and 40.65%. The results show that the proposed algorithm can improve the accuracy of normal daily load prediction compared with most single models.

In extreme weather scenarios (Table 5), XGBoost and LSTM yield similar results, with MAPE values of 6.49% and 6.73% at high temperatures and 5.30% and 5.72% at low temperatures. Due to load decomposition, VMD-TCN has the better predicted performance of 5.10% and 3.41%, respectively. Informer has errors of 6.63% and 3.22%, which perform well at low temperatures but have large short-term fluctuations at high temperatures. The MAPE of this paper's algorithm is 3.88% and 3.12% at high and low temperatures, respectively, which is 42.35%, 43.60%, 41.48%, 23.92%, and 33.79% lower compared to XGBoost, LSTM, Informer, TCN, and VMD-TCN in the high-temperature scenario, and 45.45%, 58.46%, 3.11%, 8.50%, and 26.76% lower in the low-temperature scenario. Experimental results show that the prediction accuracy of the proposed method is improved significantly in extreme weather scenarios; its effectiveness stems from decomposing load components and creating expert prediction models for different scenarios.

Table 6 compares prediction errors during holidays. XGBoost achieves MAPE of 8.29% and 3.23% on National Day and Workers' Day, respectively, performing better when only holiday components dominate. This occurs because National Day's load profile combines holiday patterns with temperature-dependent loads, whereas Workers' Day is primarily affected by holiday factors. This shows that traditional ML methods effectively model holiday characteristics but lose accuracy when adding temperature-adjustment loads. Deep learning models (LSTM, Informer, TCN) exhibit significant accuracy drops during holidays, with TCN showing the most pronounced decline (MAPE: 22.66% National Day, 19.45% Workers' Day), underperforming traditional ML methods. Notably, TCN achieves the lowest error on normal days due to its dilated convolutions, effectively capturing long-term temporal dependencies in load patterns. However, holiday loads involve complex coupled factors beyond historical trends, explaining TCN's performance gap between regular and special days. VMD-TCN improves upon standalone deep learning models but remains less effective overall. The MAPE of this paper's algorithm is 1.79% and 5.81% on Workers' Day and National Day, respectively. The prediction error is reduced by 29.92%, 40.29%, 35.19%, 74.36%, and 56.45% on the day of National Day compared to XGBoost, LSTM, Informer, TCN, VMD-TCN, and the prediction error on the day of International worker's day is reduced by 44.58%, 92.00%, 90.54%, 90.80%, 87.08%, the results show that the algorithm in this paper has high prediction accuracy in holiday scenarios. The proposed method enhances prediction performance by decomposing tasks among specialized expert models, where each focuses solely on its domain expertise. This approach mitigates overfitting and prevents limited generalizability to everyday scenarios.

In Tables 4 and 5, we observed that different models exhibit varying performance across scenarios. For instance, LSTM achieves lower MAPE during high- and low-temperature periods than normal days, while TCN performs better on weekends than on weekdays. However, the proposed algorithm consistently demonstrates superior accuracy and robustness in complex scenarios, such as extreme weather and holidays, as evidenced by its lower RMSE and MAE values. This highlights that different models may excel in specific scenarios, but the proposed algorithm maintains strong overall performance across diverse conditions.

By comparing and analyzing the prediction results of various methods in different scenarios, it can be found that the method in this paper can provide prediction results in specific scenarios, such as extreme weather and holidays, which are in line with these special load patterns, and still maintains high prediction accuracy in normal daily load prediction, with strong generalizability and robustness.

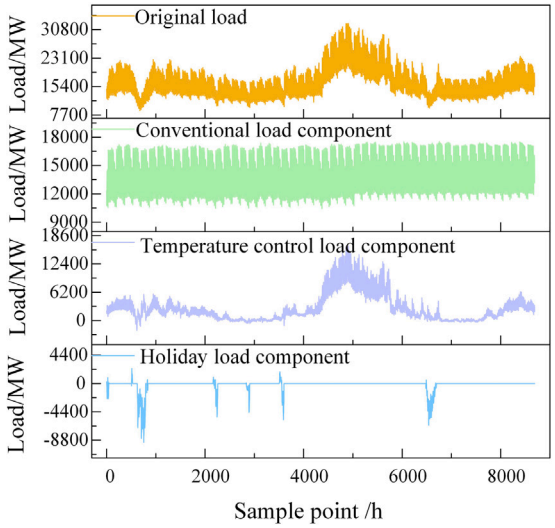


Fig. 6. The result of decomposition of each load component.

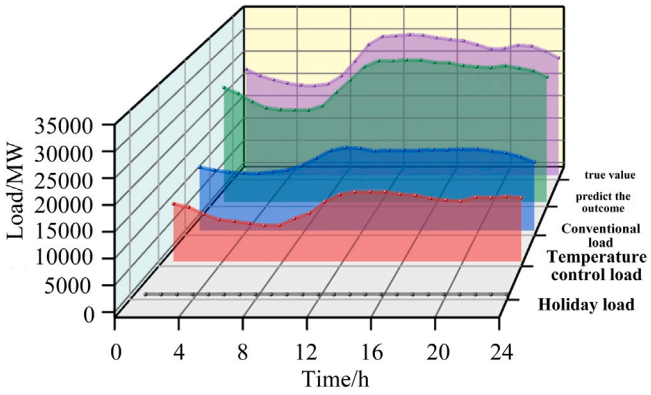


Fig. 7. Prediction curve of each load component on high-temperature day.

4.4. Interpretability analysis of predictive models

The algorithm proposed in this paper decomposes the original load into regular, temperature-regulating, and holiday load components by training on known data and predicting the unknown. As shown in Fig. 6, the interpretable decomposition results on the training set reveal that the regular load component exhibits cyclical patterns. The temperature-regulating load component is concentrated in summer and winter, with higher and more volatile values in summer. The holiday load component generally reduces loads during holidays, while load increases during shifted workdays, such as weekends adjusted to working days.

The proposed forecasting method is interpretable both in model structure and prediction results. The model structure and learning objectives are designed for interpretability prior to training, as detailed in Section 3. Additionally, the forecast results are interpretable by component. For example, Fig. 7 illustrates the predicted load components on July 26, 2019—a regular workday with no holiday load component—featuring a high proportion of temperature-regulating loads due to a maximum temperature of 39 °C and a perceived temperature of 46 °C.

As shown in Fig. 8 for each load component prediction curve on October 1, the maximum temperature of the day is 27 °C, the maximum body temperature is 29 °C, and the maximum temperature of the previous day is 30 °C, because October 1 is a legal holiday, there is a negative holiday load component and a lower share of the temperature-regulating load component.

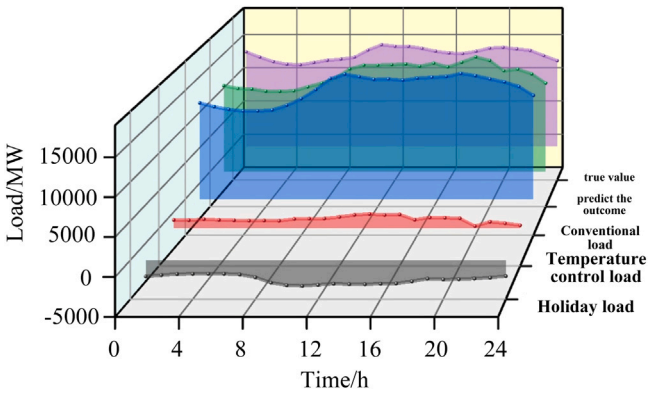


Fig. 8. Forecast curve of each load component on National Day.

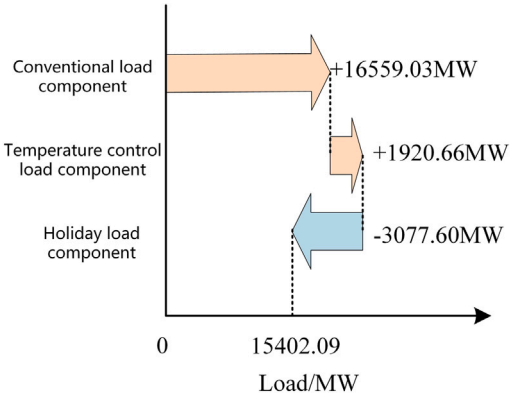


Fig. 9. Analysis of contribution degree of different load components.

As shown in Fig. 9, the load forecast at 12:00 on National Day 2019 highlights the contribution of each component, effectively explaining the model's predictions and enhancing credibility. These results support operational staff with reliable insights for decision-making.

5. Conclusion and prospect

This paper proposes an interpretable short-term load prediction method for extreme scenarios based on a mixture of experts model. Firstly, the normal daily load can be accurately and adaptively extracted by combining meteorological characteristics with curve clustering. Secondly, the load is decomposed into different components by combining prior knowledge with the expert framework. Different expert models are used to forecast, significantly improving prediction accuracy under extreme weather and holidays. Its interpretability allows for component-based load analysis, enhances model credibility, and supports power system optimization, renewable energy integration, and grid planning.

In our experiments, we adopted a fixed combination of expert models. While this approach is effective, it may not suit all tasks. Therefore, we plan to explore more flexible expert model selection strategies in future research, including the adaptive decomposition of explainable components, automated model selection, and model tuning, further to enhance the adaptability and prediction accuracy of the model. Additionally, the mixture of experts model based on explainable artificial intelligence exhibits strong interpretability and flexibility. In future work, we aim to study the deconstruction of loads into various other explainable components, such as electric vehicles and emerging loads, tailored to different load forecasting targets to achieve more refined and precise predictions.

CRediT authorship contribution statement

Yaoxian Liu: Writing – review & editing, Validation, Supervision. **Kaixin Zhang:** Writing – original draft, Software. **Songsong Chen:** Methodology, Investigation. **Xiaorui Qian:** Formal analysis. **Ying Zhou:** Visualization. **Xiangpeng Zhan:** Project administration. **Jingwen Chen:** Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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